Bank Marketing Subscription Prediction System

Academic Machine Learning Evaluation Report | Generated: 2026-10-25 23:30:53

1. Executive Summary & Problem Formulation

Direct telemarketing campaigns in the banking sector incur substantial operational expenditures while typically yielding modest conversion rates. This project develops an end-to-end Machine Learning system to classify whether prospective bank clients will subscribe to a term deposit. To enforce methodological rigor and prevent feature leakage, call duration is intentionally omitted, ensuring realistic pre-contact predictive efficacy.

2. Dataset & Experimental Setup

Parameter	Configured Value	Parameter	Configured Value
Benchmark Dataset	bank-full_clean.csv	Total Records	45,211
Training Split	36,168	Test Split	9,043
Target Variable	y ('no' / 'yes')	Positive Class (yes)	11.70% (Imbalanced)
Leakage Prevention	Excluded 'duration'	Validation Protocol	Stratified Test Partition

3. Empirical Four-Model Benchmark

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Train (s)
Logistic Regression	87.84%	47.76%	42.34%	44.89%	77.20%	5.49s
Decision Tree Classifier	88.00%	48.66%	46.41%	47.51%	77.76%	6.72s
K-Nearest Neighbours (KNN)	86.42%	41.13%	37.24%	39.09%	72.33%	10.79s
Random Forest Classifier	86.97%	44.91%	50.00%	47.32%	79.86%	135.86s

4. Technical Findings & Class Imbalance Analysis

Across the held-out evaluation subset of 9,043 instances, **Random Forest Classifier** attained the highest overall discriminative power with an ROC-AUC of 79.86%, while **Decision Tree Classifier** delivered top balanced harmonic score with an F1 of 47.51%. Due to the pronounced 88.3% / 11.7% class imbalance, accuracy alone is misleading: a naive dummy classifier predicting zero universally achieves 88.30% accuracy while failing completely on all subscribers. The balanced class-weighting framework deployed across our ensemble and linear models elevates recall for the subscription class, ensuring valuable prospect discovery for telemarketing campaign managers.

5. Literature Review & Citation Audit

Study & Authors	Venue / DOI	Verification Status & Alignment
Alessa et al. (2023)	IEEE ICCMA 10.1109/ICCMA59762.2023.10374595	**[Unverified, check before citing]** Evaluates the exact four core machine learning algorithms implemented in this project on the UCI Bank Marketing dataset.

Singh et al. (2024)	IEEE TIACOMP 10.1109/TIACOMP64125.2024.00085	**[Unverified, check before citing]** Confirms the necessity of our class-weighting strategy to safeguard recall on the minority subscription class.
Dixit & Kokila (2024)	IEEE ICONAT IEEE Document ID: 10700543	**[Unverified, check before citing]** Directly supports our comparative findings that tuned tree ensembles yield superior discriminative capability.
Tanvir et al. (2024)	arXiv / IEEE DASA 10.48550/arXiv.2410.21539	**[Verified]** Supports BankerEye's calibrated probability outputs alongside binary predictions.
Khan et al. (2021)	Springer LNCS 10.1007/978-3-030-87010-9_8	**[Unverified, check before citing]** Governs our feature importance dashboard and non-causal statistical disclaimer requirements.

6. Ethical Considerations & System Limitations

Notice: *Model predictions represent probabilistic estimates based on historical observational records. Outputs should augment human decision-making rather than act as autonomous credit or marketing rejection mechanisms. Feature correlations must not be interpreted as definitive causal proof.*